Systems of Differential Equations Reading

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In a variety of applications, systems of equations arise involving one or more functions and the derivatives of those functions. One example can be found in a simplified model of the concentration of insulin and glucose in an individual. Insulin is a hormone that reduces glucose concentrations.

Suppose that $y_1(t)$ and $y_2(t)$ give the deviation from normal of insulin and glucose concentrations, respectively. Then the rates of change $y_1'(t)$ and $y_2'(t)$ of insulin and glucose concentrations are related by

For instance, $y_2(t) = 10$ indicates a glucose level of 10 mg/dl above normal.

$$\begin{aligned} y_1'(t) &= ay_1(t) + by_2(t) \\ y_2'(t) &= cy_1(t) + dy_2(t) \end{aligned}$$

where a, b, c, and d are constants. This is an example of a system of linear differential equations. We will return to this example shortly, after developing a method for finding the solutions to such systems.

If $y = y(t)$, one of the simplest *differential equations* is

$$y' = ay \quad (1)$$

where a is a constant. Setting $y = ce^{at}$ for any constant c, we have $y' = ace^{at}$, so that our function y satisfies $y' = ay$. Our function is called a *solution* to the differential equation, and in fact, the only solutions to this differential equation have this form.

Here we assume a basic familiarity with differential equations, and so only provide a brief background.

In this lesson we describe how to find the solutions to a system of linear first-order differential equations, which has the form

$$\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + a_{13}y_3 + \cdots + a_{1n}y_n \\ y_2' &= a_{21}y_1 + a_{22}y_2 + a_{23}y_3 + \cdots + a_{2n}y_n \\ y_3' &= a_{31}y_1 + a_{32}y_2 + a_{33}y_3 + \cdots + a_{3n}y_n \\ &\vdots \qquad\qquad \vdots \qquad\qquad \vdots \qquad\qquad \vdots \\ y_n' &= a_{n1}y_1 + a_{n2}y_2 + a_{n3}y_3 + \cdots + a_{nn}y_n \end{aligned} \quad (2)$$

We assume that $y_1 = y_1(t), \ldots, y_n = y_n(t)$ are each differentiable functions. The system is linear because the functions are linearly related, and it is first-order because the derivatives that appear are first order only. If we denote

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{y}' = \begin{bmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{bmatrix}, \quad \text{and} \quad A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

then the system (2) can be expressed compactly as $\mathbf{y}' = A\mathbf{y}$. This matrix equation resembles the differential equation (1), which suggests that a solution to our system might have the form

$$\mathbf{y} = \begin{bmatrix} u_1 e^{\lambda t} \\ u_2 e^{\lambda t} \\ \vdots \\ u_n e^{\lambda t} \end{bmatrix} = e^{\lambda t}\mathbf{u}$$

where $\mathbf{u}$ is a constant vector. If $\mathbf{y}$ is so defined, then $\mathbf{y}' = \lambda e^{\lambda t}\mathbf{u} = \lambda(e^{\lambda t}\mathbf{u})$ and $A\mathbf{y} = A(e^{\lambda t}\mathbf{u}) = e^{\lambda t}(A\mathbf{u})$. Since $e^{\lambda t} \neq 0$, this tells us that is a solution to $\mathbf{y}' = A\mathbf{y}$ exactly when

$$A\mathbf{u} = \lambda\mathbf{u}$$

That is, $\mathbf{y} = e^{\lambda t}\mathbf{u}$ is a solution when λ is an eigenvalue of A with associated eigenvector $\mathbf{u}$.

In many cases, an $n \times n$ matrix A will have n linearly independent eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ with associated eigenvalues $\lambda_1, \ldots, \lambda_n$. If we form the linear combination

$$\mathbf{y} = c_1 e^{\lambda_1 t}\mathbf{u}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{u}_n$$

where $c_1, \ldots, c_n$ are constants, then $\mathbf{y}' = c_1\lambda_1 e^{\lambda_1 t}\mathbf{u}_1 + \cdots + c_n\lambda_n e^{\lambda_n t}\mathbf{u}_n$ and

$$\begin{aligned} A\mathbf{y} &= A\left(c_1 e^{\lambda_1 t}\mathbf{u}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{u}_n\right) \\ &= c_1 e^{\lambda_1 t}A\mathbf{u}_1 + \cdots + c_n e^{\lambda_n t}A\mathbf{u}_n \\ &= c_1\lambda_1 e^{\lambda_1 t}\mathbf{u}_1 + \cdots + c_n\lambda_n e^{\lambda_n t}\mathbf{u}_n = \mathbf{y}' \end{aligned}$$

Thus $\mathbf{y}$ is a solution to $\mathbf{y}' = A\mathbf{y}$. It turns out that all solutions to this type of system of differential equations will have this form. The set of all solutions is called the **general solution** for the system. This is summarized in the next theorem.

THEOREM 15

Suppose that $\mathbf{y}' = A\mathbf{y}$ is a first-order linear system of differential equations. If A is an $n \times n$ diagonalizable matrix, then the general solution to the system is given by

$$\mathbf{y} = c_1 e^{\lambda_1 t}\mathbf{u}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{u}_n$$

where $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are n linearly independent eigenvectors with associated eigenvalues $\lambda_1, \ldots, \lambda_n$, and $c_1, \ldots, c_n$ are constants.

Note that if A is diagonalizable, then there must be n linearly independent eigenvectors. Also, the eigenvalues may be repeated to reflect multiplicities.

Example 1

The concentrations of insulin and glucose in an individual interact with each other and vary over time. A mathematical model for the concentrations of insulin and glucose in an individual is given by the system

$$\begin{aligned} y_1' &= -0.05y_1 + 0.225y_2 \\ y_2' &= -0.3y_1 - 0.65y_2 \end{aligned}$$

Find the general solution for this system.

Solution

The coefficient matrix is $A = \begin{bmatrix} -0.05 & 0.225 \\ -0.3 & -0.65 \end{bmatrix}$. Applying our standard methods, we find that the eigenvalues and eigenvectors are

$$\lambda_1 = -0.5 \quad \Rightarrow \quad \mathbf{u}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \lambda_2 = -0.2 \quad \Rightarrow \quad \mathbf{u}_2 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

By Theorem 15, the general solution is

$$\mathbf{y} = c_1 e^{-0.5t}\mathbf{u}_1 + c_2 e^{-0.2t}\mathbf{u}_2 = c_1 e^{-0.5t}\begin{bmatrix} -1 \\ 2 \end{bmatrix} + c_2 e^{-0.2t}\begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

The functions giving the insulin and glucose concentrations are

$$\begin{aligned} y_1 &= -c_1 e^{-0.5t} + 3c_2 e^{-0.2t} \\ y_2 &= 2c_1 e^{-0.5t} - 2c_2 e^{-0.2t} \end{aligned}$$

Arms Races

After the end of World War I, Lewis F. Richardson (who pioneered the use of mathematics in meteorology) proposed a model to describe the evolution of an arms race between two countries. Here we consider a simplified version of this model. Let $y_1 = y_1(t)$ and $y_2 = y_2(t)$ denote the quantity of arms held by two different nations. The derivatives y_1' and y_2' represent the rate of change in the size of each nation's arsenal. Each nation is concerned about security against the other and acquires arms in proportion to those held by its opponent. There is also a cost of acquiring arms, which tends to reduce the rate of additions to each nation's arsenal in proportion to arsenal size. These factors are incorporated into the system of differential equations

$$\begin{aligned} y_1' &= -dy_1 + ey_2 \\ y_2' &= fy_1 - gy_2 \end{aligned} \quad (3)$$

The constants d, e, f, and g are all positive and depend on the particular situation, with e and f dictated by the degree of fear that each country has of the other and d and g the level of aversion to additional spending on arms in each country.

Example 2

Suppose that we have two countries in an arms race modeled by the system of differential equations

$$\begin{aligned} y_1' &= -3y_1 + 2y_2 \\ y_2' &= 3y_1 - 2y_2 \end{aligned}$$

Find the general solution for this system. Then find the formula for $y_1(t)$ and $y_2(t)$ if $y_1(0) = 5$ and $y_2(0) = 15$.

Solution

The coefficient matrix for this system is $A = \begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix}$. Applying our usual methods for finding the eigenvalues and eigenvectors, we have

$$\lambda_1 = -5 \quad \Rightarrow \quad \mathbf{u}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \lambda_2 = 0 \quad \Rightarrow \quad \mathbf{u}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Therefore, by Theorem 15, the general solution for this system is

$$\mathbf{y} = c_1 e^{-5t}\mathbf{u}_1 + c_2 e^{0}\mathbf{u}_2 = c_1 e^{-5t}\begin{bmatrix} -1 \\ 1 \end{bmatrix} + c_2\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Extracting the two functions y_1 and y_2, we have

$$\begin{aligned} y_1 &= -c_1 e^{-5t} + 2c_2 \\ y_2 &= c_1 e^{-5t} + 3c_2 \end{aligned}$$

Evaluating these functions at $t = 0$ and using the equations $y_1(0) = 5$ and $y_2(0) = 15$ yield the system

$$\begin{aligned} -c_1 + 2c_2 &= 5 \\ c_1 + 3c_2 &= 15 \end{aligned}$$

This linear system has unique solution $c_1 = 3$ and $c_2 = 4$. Hence

$$\begin{aligned} y_1 &= -3e^{-5t} + 8 \\ y_2 &= 3e^{-5t} + 12 \end{aligned}$$

The next example shows what we do if the coefficient matrix for a first-order linear system has repeated real eigenvalues.

Example 3

Find the general solution for the system

$$\begin{aligned} y_1' &= 2y_1 + y_2 - y_3 \\ y_2' &= 2y_1 + 3y_2 - 2y_3 \\ y_3' &= -3y_1 - 3y_2 + 4y_3 \end{aligned}$$

Solution

Here the coefficient matrix is given by

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 2 & 3 & -2 \\ -3 & -3 & 4 \end{bmatrix}$$

The characteristic polynomial for A is $-(\lambda - 7)(\lambda - 1)^2$, and bases for the eigenspaces are

$$\lambda_1 = 7 \quad \Rightarrow \quad \text{Basis: } \left\{\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}\right\}, \quad \lambda_2 = 1 \quad \Rightarrow \quad \text{Basis: } \left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}\right\}$$

Although $\lambda_2 = 1$ has multiplicity 2, since the eigenspace also has dimension 2, Theorem 15 still applies. (The case where an eigenspace has less than maximal dimension is more complicated and not included here.) We just repeat the term corresponding to $\lambda_2 = 1$ twice, once for each of the linearly independent eigenvectors. The general solution is

$$\mathbf{y} = c_1 e^{7t}\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} + c_2 e^{t}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_3 e^{t}\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Writing the individual functions, we have

$$\begin{aligned} y_1 &= c_1 e^{7t} + (c_2 + c_3)\, e^{t} \\ y_2 &= 2c_1 e^{7t} - c_3 e^{t} \\ y_3 &= -3c_1 e^{7t} + c_2 e^{t} \end{aligned}$$

Complex Eigenvalues

Suppose that our system of linear differential equations has a real coefficient matrix A with complex eigenvalues. We can express the general solution just as we did with real eigenvalues. However, when λ and associated eigenvector $\mathbf{u}$ are both complex, the product $e^{\lambda t}\mathbf{u}$ typically is as well. It is generally preferable to have solutions free of complex terms, and with a bit of extra thought we can.

To get us started, we recall a few properties of complex numbers. (All are given in Lesson 3, Complex Eigenvalues and Eigenvectors.)

- $e^{a+ib} = e^a(\cos(b) + i\sin(b))$
- $\frac{1}{2}(\overline{z} + z) = \mathrm{Re}(z)$
- $e^{\overline{z}} = \overline{e^z}$
- $\frac{1}{2i}(\overline{z} - z) = \mathrm{Im}(z)$

Now let A be a real matrix with complex eigenvalue $\lambda = \alpha + i\beta$ and associated eigenvector $\mathbf{u}$. Then $\overline{\lambda} = \alpha - i\beta$ is also an eigenvalue and has associated eigenvector $\overline{\mathbf{u}}$. Instead of taking linear combinations of $e^{\lambda t}\mathbf{u}$ and $e^{\overline{\lambda} t}\overline{\mathbf{u}}$ for the general solution, we take linear combinations of

$$\begin{aligned} \mathbf{y}_1 &= \frac{1}{2}\left(e^{\overline{\lambda} t}\overline{\mathbf{u}} + e^{\lambda t}\mathbf{u}\right) = \mathrm{Re}\left(e^{\lambda t}\mathbf{u}\right) \\ \mathbf{y}_2 &= \frac{1}{2i}\left(e^{\overline{\lambda} t}\overline{\mathbf{u}} + e^{\lambda t}\mathbf{u}\right) == \mathrm{Im}\left(e^{\lambda t}\mathbf{u}\right) \end{aligned}$$

Note that both $\mathrm{Re}(e^{\lambda t}\mathbf{u})$ and $\mathrm{Im}(e^{\lambda t}\mathbf{u})$ are real-valued, so that the general solution will be made up of real-valued functions. To find $\mathrm{Re}(e^{\lambda t}\mathbf{u})$ and $\mathrm{Im}(e^{\lambda t}\mathbf{u})$, we compute

$$\begin{aligned} e^{\lambda t}\mathbf{u} &= e^{\alpha t + i\beta t}\mathbf{u} \\ &= e^{\alpha t}\left(\cos\left(\beta t\right) + i\sin\left(\beta t\right)\right)\left(\mathrm{Re}\left(\mathbf{u}\right) + i\,\mathrm{Im}\left(\mathbf{u}\right)\right) \\ &= e^{\alpha t}\left(\cos\left(\beta t\right)\mathrm{Re}(B) - \sin\left(\beta t\right)\mathrm{Im}\left(\mathbf{u}\right)\right) + ie^{\alpha t}\left(\sin\left(\beta t\right)\mathrm{Re}\left(\mathbf{u}\right) + \cos\left(\beta t\right)\mathrm{Im}\left(\mathbf{u}\right)\right) \end{aligned}$$

Separating the real and imaginary parts, we have

$$\begin{aligned} \mathbf{y}_1 &= e^{\alpha t}\left(\cos\left(\beta t\right)\mathrm{Re}\left(\mathbf{u}\right) - \sin\left(\beta t\right)\mathrm{Im}\left(\mathbf{u}\right)\right) \\ \mathbf{y}_2 &= e^{\alpha t}\left(\sin\left(\beta t\right)\mathrm{Re}\left(\mathbf{u}\right) + \cos\left(\beta t\right)\mathrm{Im}\left(\mathbf{u}\right)\right) \end{aligned} \quad (4)$$

Example 4

Find the general solution for the system

$$\begin{aligned} y_1' &= 6y_1 - 5y_1 \\ y_2' &= 5y_1 - 2y_2 \end{aligned}$$

Solution

Here the coefficient matrix is

$$A = \begin{bmatrix} 6 & -5 \\ 5 & -2 \end{bmatrix}$$

The characteristic polynomial for this matrix is $\det(A - \lambda I_2) = \lambda^2 - 4\lambda + 13$. Applying the quadratic formula and the usual matrix manipulations yields the eigenvalues and eigenvectors

$$\lambda_1 = 2 + 3i \quad \Rightarrow \quad \mathbf{u}_1 = \begin{bmatrix} 4 + 3i \\ 5 \end{bmatrix}, \quad \lambda_2 = 2 - 3i \quad \Rightarrow \quad \mathbf{u}_1 = \begin{bmatrix} 4 - 3i \\ 5 \end{bmatrix}$$

The two eigenvalues are a complex conjugate pair, so we use the formulas for $\mathbf{y}_1$ and $\mathbf{y}_2$ given in (4) to find

$$\begin{aligned} \mathbf{y}_1 &= e^{2t}\left(\cos\left(3t\right)\mathrm{Re}\left(\mathbf{u}_1\right) - \sin\left(3t\right)\mathrm{Im}\left(\mathbf{u}_1\right)\right) = e^{2t}\left(\cos\left(3t\right)\begin{bmatrix} 4 \\ 5 \end{bmatrix} - \sin\left(3t\right)\begin{bmatrix} 3 \\ 0 \end{bmatrix}\right) \\ \mathbf{y}_2 &= e^{2t}\left(\sin\left(3t\right)\mathrm{Re}\left(\mathbf{u}_1\right) + \cos\left(3t\right)\mathrm{Im}\left(\mathbf{u}_1\right)\right) = e^{2t}\left(\sin\left(3t\right)\begin{bmatrix} 4 \\ 5 \end{bmatrix} + \cos\left(3t\right)\begin{bmatrix} 3 \\ 0 \end{bmatrix}\right) \end{aligned}$$

Hence the general solution is

$$\begin{aligned} \mathbf{y} &= c_1\mathbf{y}_1 + c_2\mathbf{y}_2 \\ &= c_1 e^{2t}\left(\cos\left(3t\right)\begin{bmatrix} 4 \\ 5 \end{bmatrix} - \sin\left(3t\right)\begin{bmatrix} 3 \\ 0 \end{bmatrix}\right) + c_2 e^{2t}\left(\sin\left(3t\right)\begin{bmatrix} 4 \\ 5 \end{bmatrix} + \cos\left(3t\right)\begin{bmatrix} 3 \\ 0 \end{bmatrix}\right) \end{aligned}$$

The individual functions are

$$\begin{aligned} y_1 &= \left((4c_1 + 3c_2)\cos\left(3t\right) + \left(4c_2 - 3c_1\right)\sin\left(3t\right)\right)e^{2t} \\ y_2 &= \left(5c_1\cos\left(3t\right) + 5c_2\sin\left(3t\right)\right)e^{2t} \end{aligned}$$

The next example features a system with a combination of real and complex eigenvalues.

Example 5

Find the general solution for the system

$$\begin{aligned} y_1' &= -10y_1 + 6y_2 - 3y_3 \\ y_2' &= -12y_1 + 6y_2 - 5y_3 \\ y_3' &= 8y_1 - 4y_2 + 3y_3 \end{aligned}$$

Solution

The coefficient matrix is

$$A = \begin{bmatrix} -10 & 6 & -3 \\ -12 & 6 & -5 \\ 8 & -4 & 3 \end{bmatrix}$$

and the characteristic polynomial is

$$\det\left(A - \lambda I_3\right) = -\lambda^3 - \lambda^2 - 4\lambda - 4 = -\lambda^2\left(\lambda + 1\right) - 4\left(\lambda + 1\right) = -\left(\lambda^2 + 4\right)\left(\lambda + 1\right)$$

giving us eigenvalues $\lambda = -1, \pm 2i$. The associated eigenvectors are

$$\lambda_1 = -1 \Rightarrow \mathbf{u}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad \lambda_2 = 2i \Rightarrow \mathbf{u}_2 = \begin{bmatrix} 9 - 3i \\ 12 - 2i \\ -8 \end{bmatrix}, \quad \lambda_3 = -2i \Rightarrow \mathbf{u}_3 = \begin{bmatrix} 9 + 3i \\ 12 + 2i \\ -8 \end{bmatrix}$$

We treat the real and complex eigenvalues separately and form a linear combination of the components at the end.
For $\lambda_1 = -1$, we have

$$\mathbf{y}_1 = e^{-t}\mathbf{u}_1 = e^{-t}\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

For the complex conjugate pair $\pm 2i$, we have

$$\begin{aligned} \mathbf{y}_2 &= e^{0}\left(\cos\left(2t\right)Re\left(\mathbf{u}_2\right) - \sin\left(2t\right)Im\left(\mathbf{u}_2\right)\right) = \cos\left(2t\right)\begin{bmatrix} 9 \\ 12 \\ -8 \end{bmatrix} - \sin\left(2t\right)\begin{bmatrix} -3 \\ -2 \\ 0 \end{bmatrix} \\ \mathbf{y}_3 &= e^{0}\left(\sin\left(2t\right)Re\left(\mathbf{u}_2\right) + \cos\left(2t\right)Im\left(\mathbf{u}_2\right)\right) = \sin\left(2t\right)\begin{bmatrix} 9 \\ 12 \\ -8 \end{bmatrix} + \cos\left(2t\right)\begin{bmatrix} -3 \\ -2 \\ 0 \end{bmatrix} \end{aligned}$$

The general solution is then

$$\mathbf{y} = c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + c_3\mathbf{y}_3$$

for $\mathbf{y}_1$, $\mathbf{y}_2$, and $\mathbf{y}_3$ above.

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